DL FOR MUSIC GENERATION AND ANALYSIS HW0

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Source Code and Model Link

OUTLINE

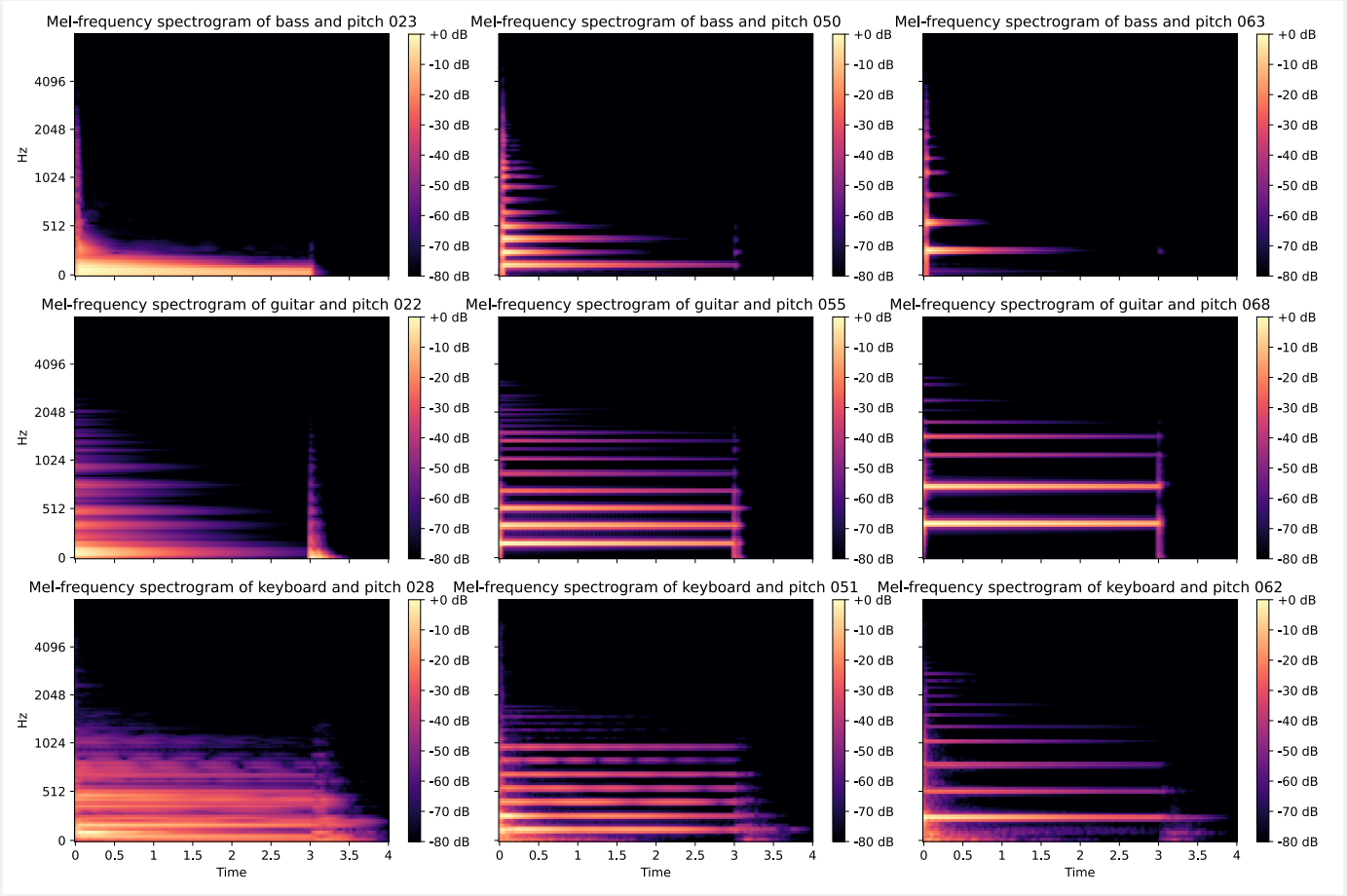
- **Task 1 – Mel-Spectrogram Visualization**
  - Settings
  - Outcome
  - Observations
- **Task 2 – Traditional ML Model**
  - Settings
  - Outcome
- **Task 3 – DL Model**
  - Settings
  - Outcome – with / without taking logarithm on mel-spectrogram

TASK 1 - SETTING

- Setting
  - Instruments: bass, guitar and keyboard
  - Dataset: data in the testing dataset.
  - Pitches:
    - roughly categorized the pitches as low (~30), medium (30~60) and high (60~) and select one from each of the three sets.
    - In order to have an effective comparison of the pitches across the instruments, I try to select as closer pitch value as possible for the three different instruments.

TASK 1 – OUTCOME

- The outcome is shown in the below figure.
- From the nine graphs, we can analyze from crossing the pitches (along the rows) and the instruments (along the columns).



TASK 1 – OBSERVATIONS

- **Observation 1:** When the pitch increases, whatever the instrument is, the frequency will concentrate in a higher range.
  - When the pitch is higher, the fundamental frequency as well as the overtones (i.e., the frequencies higher than the fundamental frequency) are higher.
  - Thus, the cluster will appear in a higher value range in frequency domain
- **Observation 2:** The bin-patterns in the picture of the lowest pitch are not as clear as those of the other two pitches.
  - Lower pitches often have waveforms that are less distinct and more spread out over time compared to higher frequencies.
  - This spread can make distinct harmonics and overtones less apparent in a spectrogram

TASK 1 – OBSERVATIONS

- **Observation 3:** we can see that whatever the pitches of bass, the frequencies are decay much faster than the other two instruments.
  - I think it reflects the “difference of timbres” of the different instruments.

TASK 2 – SETTING

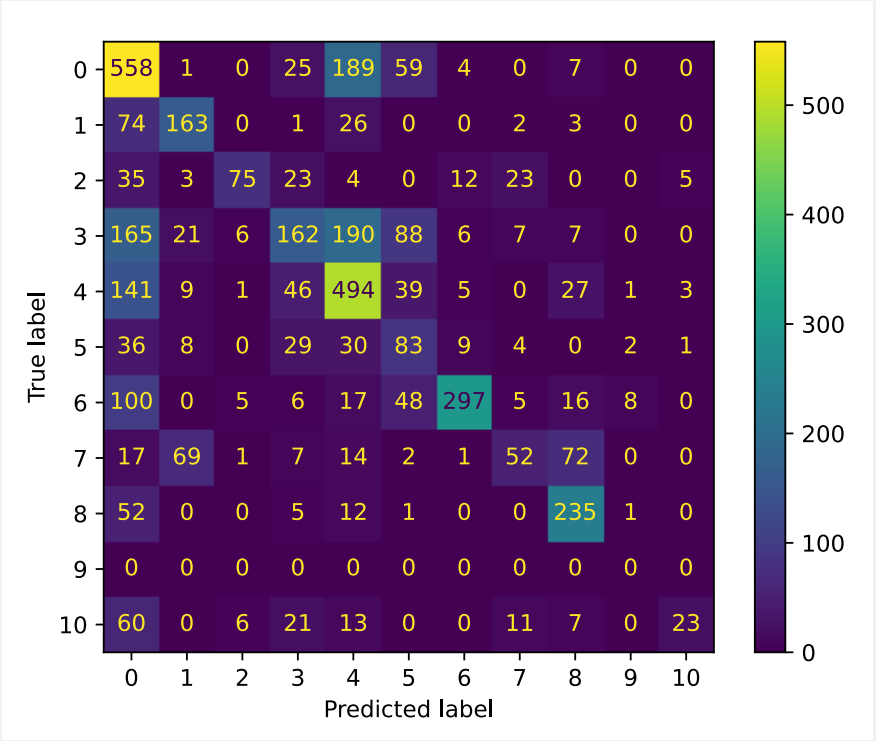
- Model Setting
  - Model: k-Neighborhood Nearest (kNN)
  - Hyper parameter: k (# of neighbor points)
  - Search range: 1, 10, 20, 30, 40, 50
- Feature Extraction
  - FFT window size: 2048 / hop length: 512
  - I transform the data into Mel-frequency cepstral coefficient (MFCC) with 20 coefficients and taking average over time domain to reduce the dimensionality
- Data Preprocessing
  - standardize with the dataset mean and standard deviation

TASK 2 – OUTCOME

- Best Hyperparameter
  - The best number of k turned out to be 50, which is the largest value of the searching range.
- Accuracy
  - Top 1: 52.17%
  - Top 3: 81.10%

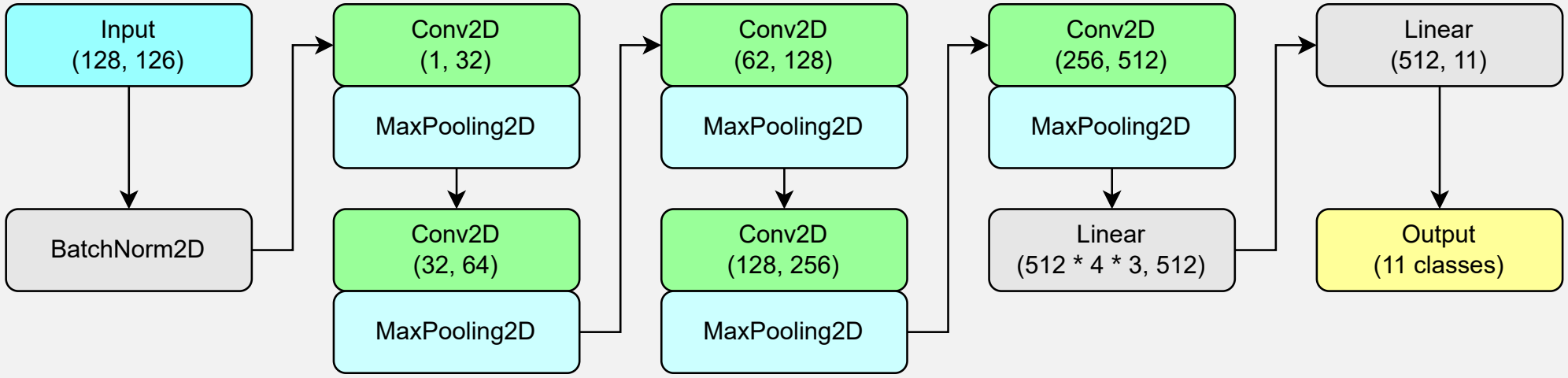
TASK 2 – OUTCOME

- Confusion Matrix
  - The model has better performance on class 0, class 4, class 6 and class 8.
  - Model seems to have a higher possibility to classify the other 10 classes as class 0, which may due to the data imbalance issue of the dataset.



TASK 3 – SETTING

- Model Structure
  - 2D - Convolution: kernel size = 3 * 3 / stride = 1, 1 / padding = 1, 1
  - MaxPooling: kernel size = 3 * 3 / stride = 1, 1 / padding = 0
  - Activation: RELU

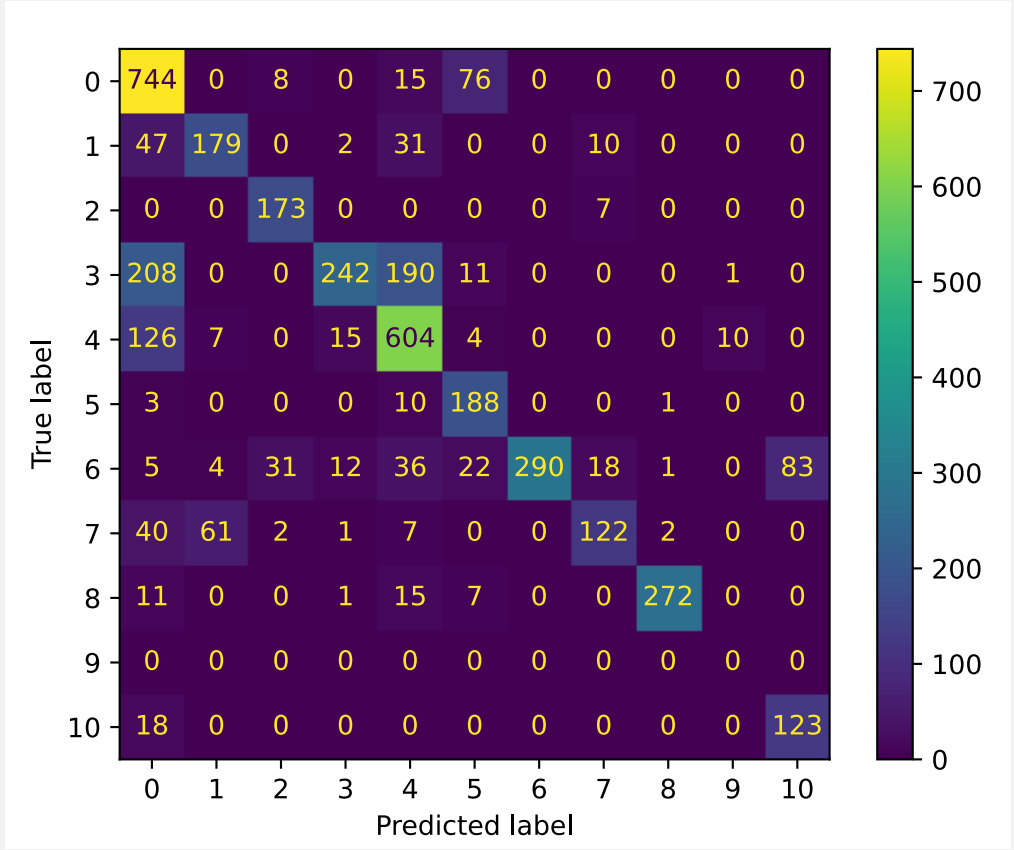


TASK 3 – SETTING

- Data Loading
  - FFT window size: 2048 / hop length: 512
  - Data shape: (128, 126)
- Training Configuration
  - # of epochs: 100
  - Batch size: 1024
  - Learning rate: 0.001
  - Early stop: 50 epochs without validation loss decreasing
  - Optimizer: Adam (with default settings)
  - Loss function: CrossEntropy

TASK 3 – OUTCOME WITH LOGARITHM

- Accuracy
  - Top 1: 71.70%
  - Top 3: 90.04%
- Confusion Matrix
  - The model has better performance than kNN, with about 17% leading.
  - Still, this model has higher possibility to classify the other classes as class 0.



TASK 3 – OUTCOME W/O LOGARITHM

- Accuracy
  - Top 1: 54.96%
  - Top 3: 82.74%
- Confusion Matrix
  - comparing to the one with logarithm mel-spectrogram, the model is drastically underperform.
  - The main factor is that when taking logarithm, the input value will be transform into a smaller range, and thus remove the extreme values or outliers.

